Spectral Simulation Report

Automated Analysis

June 19, 2025

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1 Sample: 1286

1.1 Simulation Parameters

Parameter	Value
Temperature (T)	6313 K
Electron Density (n_e)	2.36×10^{15} cm^{-3}
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	35.70

Table 1: Simulation parameters and additional metrics for sample 1286.

1.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	0.29	132.0
Ar	5.24	377.0
C	12.98	207.0
Ca	3.17	150.0
Cl	3.71	12.0
Co	12.05	983.0
Cr	7.04	397.0
Fe	11.85	2301.0
Mg	10.75	190.0
Mn	0.54	642.0
N	2.25	396.0
Na	9.24	326.0
Ni	1.22	478.0
O	1.46	10.0
S	10.78	25.0
Si	0.77	182.0
Ti	6.67	1181.0
background	N/A	797.0
Total Positions		8786.0

Table 2: Composition analysis for sample 1286.

1.3 Spectral Plot

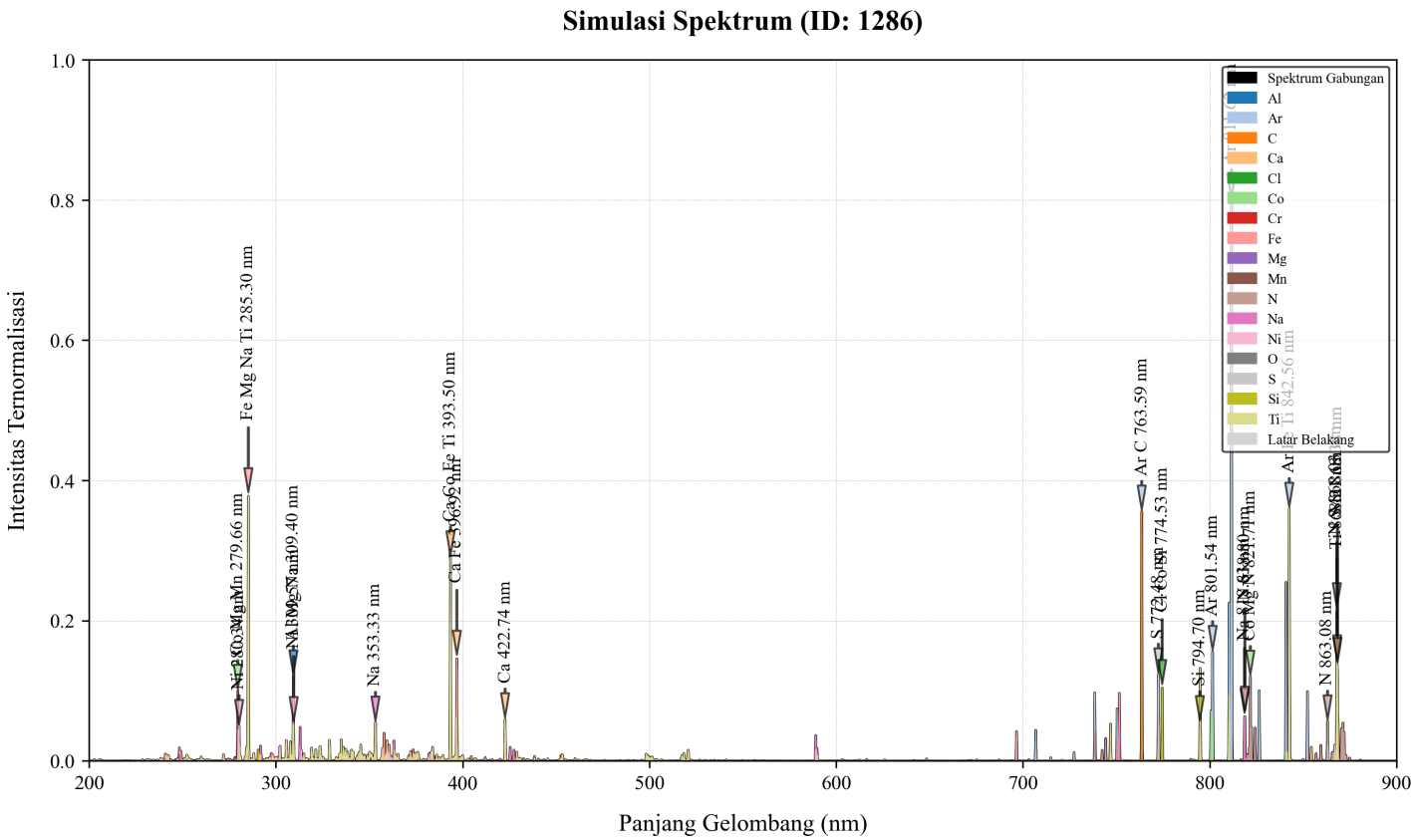

Figure 1: Spectral simulation plot for sample 1286 with multi-label encoding and detailed composition analysis.

2 Sample: 507

2.1 Simulation Parameters

Parameter	Value
Temperature (T)	6818 K
Electron Density (n_e)	2.60×10^{15} cm^{-3}
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	39.85

Table 3: Simulation parameters and additional metrics for sample 507.

2.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	11.76	202.0
Ar	13.81	592.0
C	0.00	0.0
Ca	5.30	150.0
Cl	10.16	12.0
Co	0.00	0.0
Cr	13.05	397.0
Fe	0.00	0.0
Mg	1.09	186.0
Mn	7.87	696.0
N	0.00	0.0
Na	7.21	325.0
Ni	12.23	478.0
O	9.95	66.0
S	4.89	25.0
Si	2.68	206.0
Ti	0.00	0.0
background	N/A	1900.0
Total Positions		5235.0

Table 4: Composition analysis for sample 507.

2.3 Spectral Plot

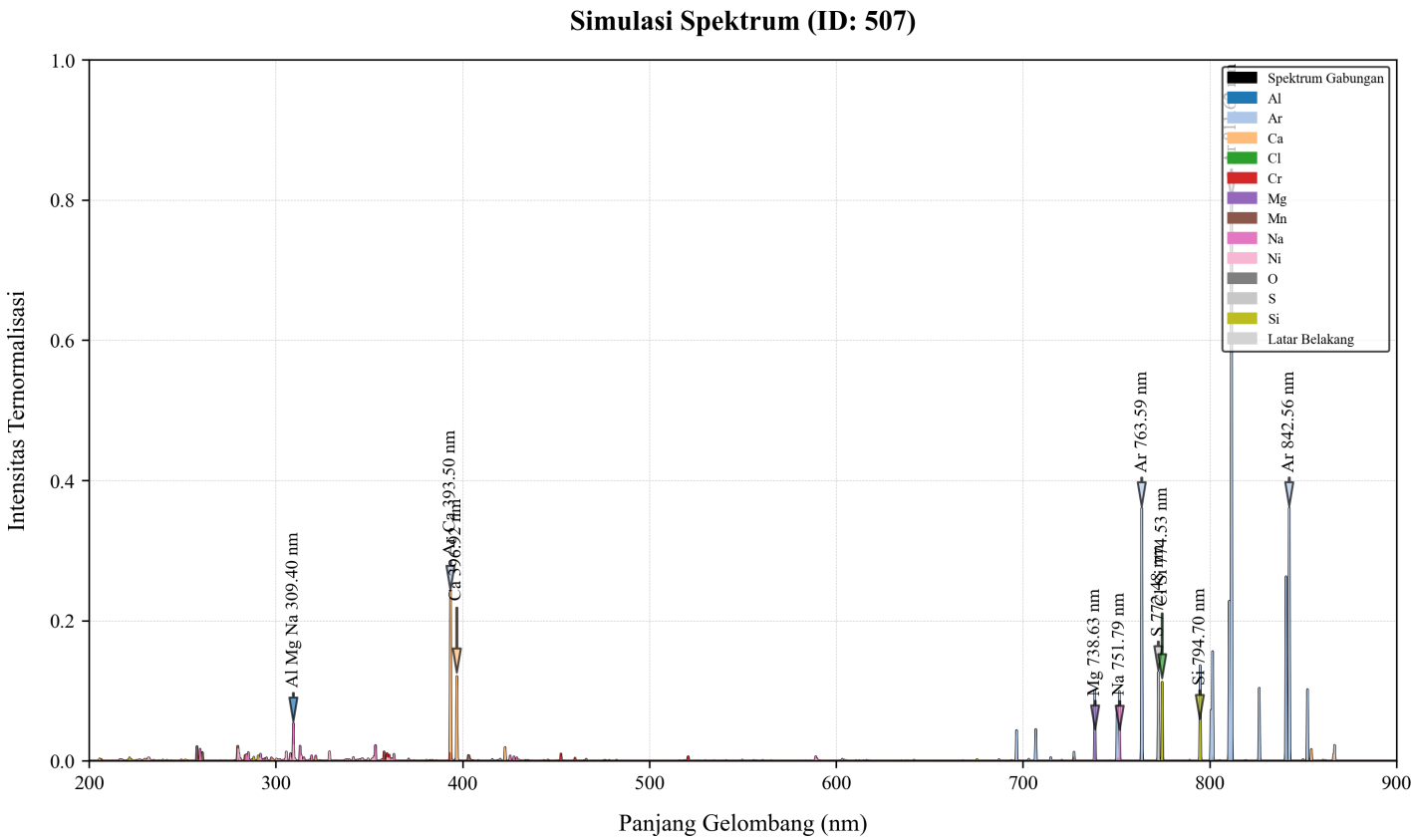

Figure 2: Spectral simulation plot for sample 507 with multi-label encoding and detailed composition analysis.

3 Sample: 16160

3.1 Simulation Parameters

Parameter	Value
Temperature (T)	14 899 K
Electron Density (n_e)	1.00×10^{18} cm^{-3}
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	42.44

Table 5: Simulation parameters and additional metrics for sample 16160.

3.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	15.82	462.0
Ar	0.00	0.0
C	0.00	0.0
Ca	21.44	150.0
Cl	14.66	35.0
Co	0.00	0.0
Cr	7.40	397.0
Fe	0.00	0.0
Mg	6.67	358.0
Mn	3.85	706.0
N	0.00	0.0
Na	2.27	330.0
Ni	18.67	478.0
O	0.00	0.0
S	1.65	228.0
Si	7.59	223.0
Ti	0.00	0.0
background	N/A	1837.0
Total Positions		5204.0

Table 6: Composition analysis for sample 16160.

3.3 Spectral Plot

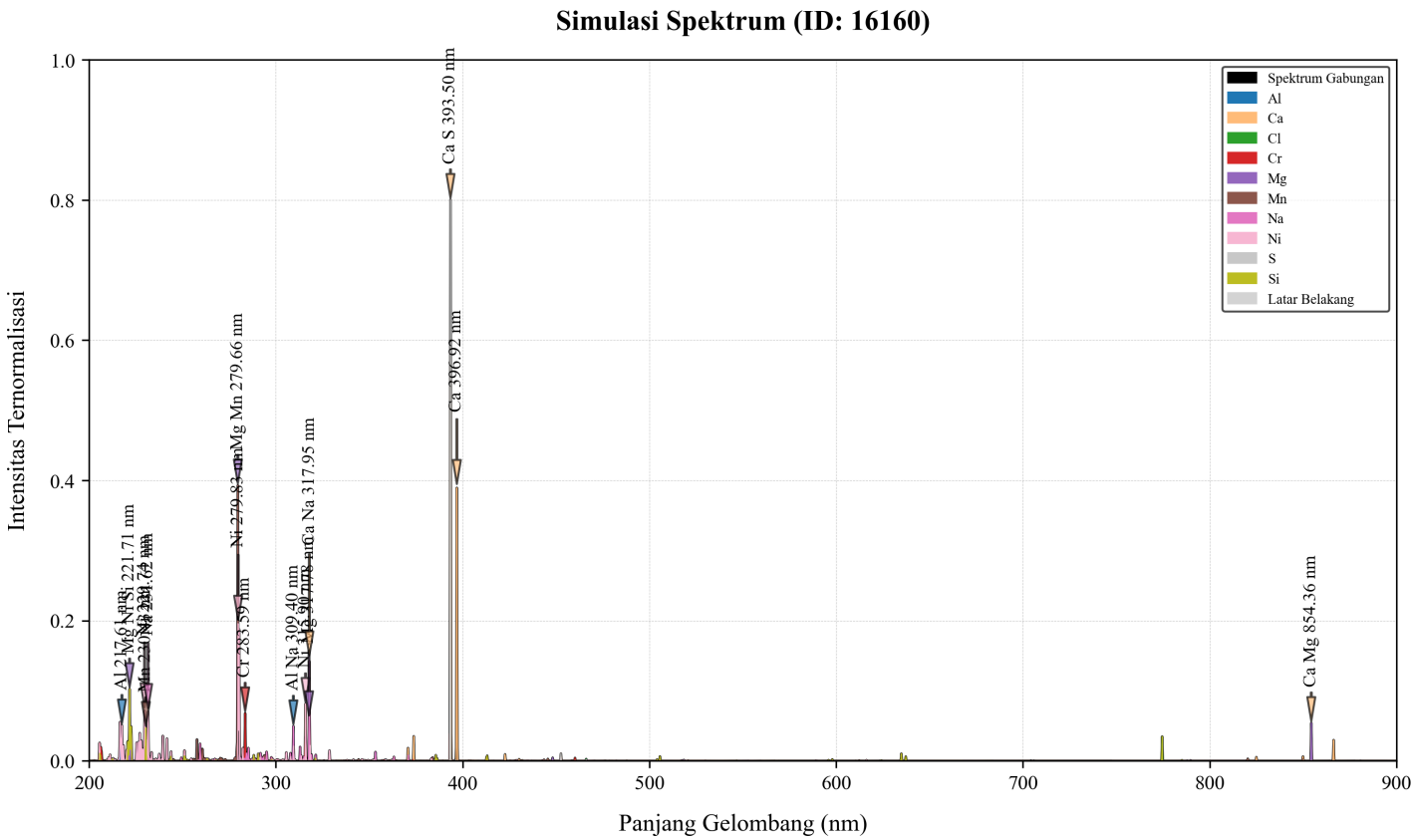

Figure 3: Spectral simulation plot for sample 16160 with multi-label encoding and detailed composition analysis.

4 Sample: 12253

4.1 Simulation Parameters

Parameter	Value
Temperature (T)	11 162 K
Electron Density (n_e)	1.00×10^{14} cm^{-3}
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	41.84

Table 7: Simulation parameters and additional metrics for sample 12253.

4.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	1.95	459.0
Ar	8.30	727.0
C	2.85	322.0
Ca	3.82	150.0
Cl	5.67	35.0
Co	1.89	1402.0
Cr	9.94	397.0
Fe	0.00	0.0
Mg	9.33	350.0
Mn	8.76	705.0
N	10.48	528.0
Na	4.06	316.0
Ni	1.39	466.0
O	5.63	167.0
S	8.53	225.0
Si	10.08	223.0
Ti	7.33	1188.0
background	N/A	713.0
Total Positions		8373.0

Table 8: Composition analysis for sample 12253.

4.3 Spectral Plot

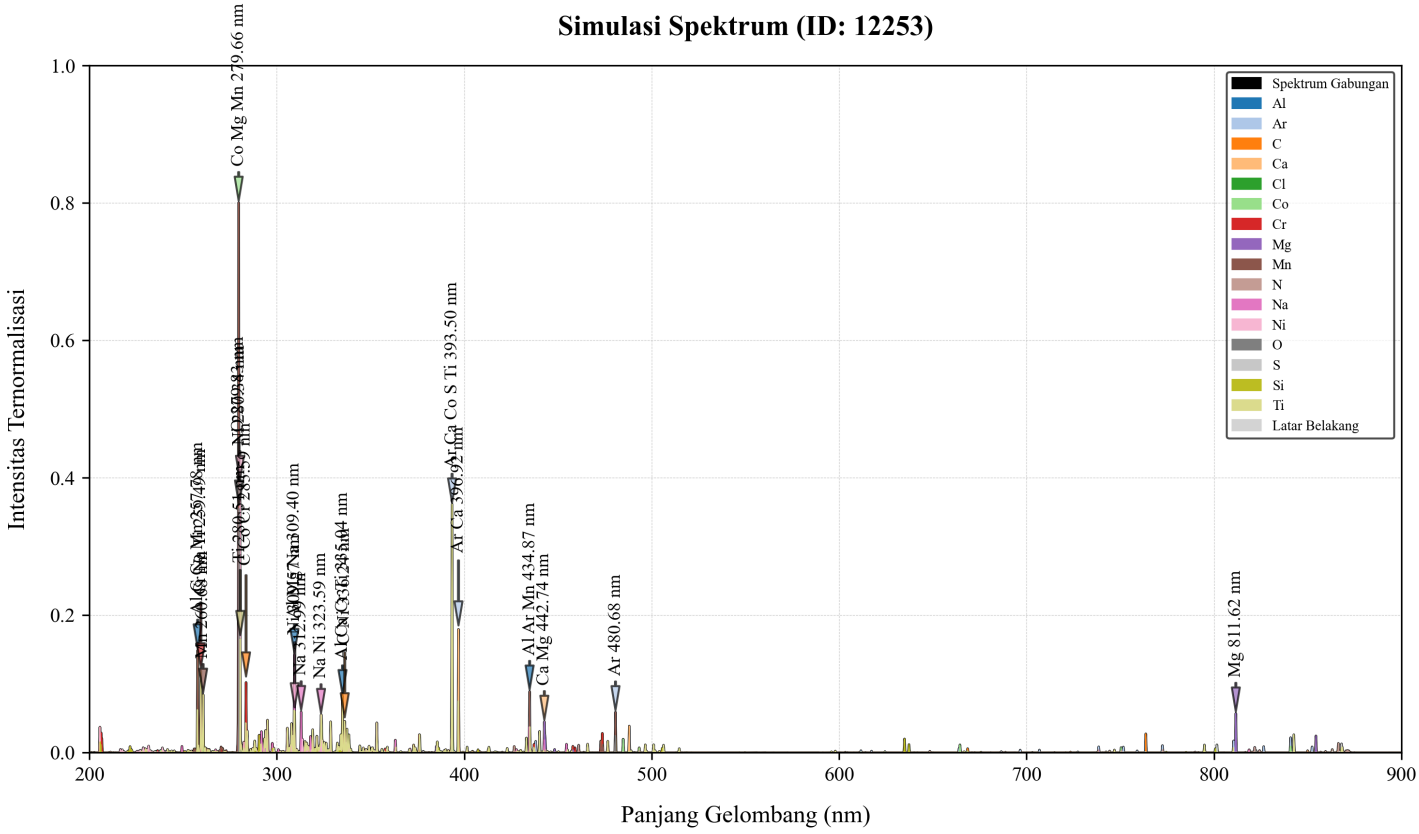

Figure 4: Spectral simulation plot for sample 12253 with multi-label encoding and detailed composition analysis.

5 Sample: 3019

5.1 Simulation Parameters

Parameter	Value
Temperature (T)	6313 K
Electron Density (n_e)	5.59×10^{16} cm^{-3}
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	36.09

Table 9: Simulation parameters and additional metrics for sample 3019.

5.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	7.74	153.0
Ar	5.49	266.0
C	5.22	187.0
Ca	7.97	150.0
Cl	6.70	12.0
Co	8.25	850.0
Cr	1.44	397.0
Fe	2.40	2132.0
Mg	6.91	163.0
Mn	7.83	646.0
N	6.19	396.0
Na	3.75	330.0
Ni	1.16	478.0
O	1.27	10.0
S	9.98	25.0
Si	9.33	176.0
Ti	8.37	1181.0
background	N/A	908.0
Total Positions		8460.0

Table 10: Composition analysis for sample 3019.

5.3 Spectral Plot

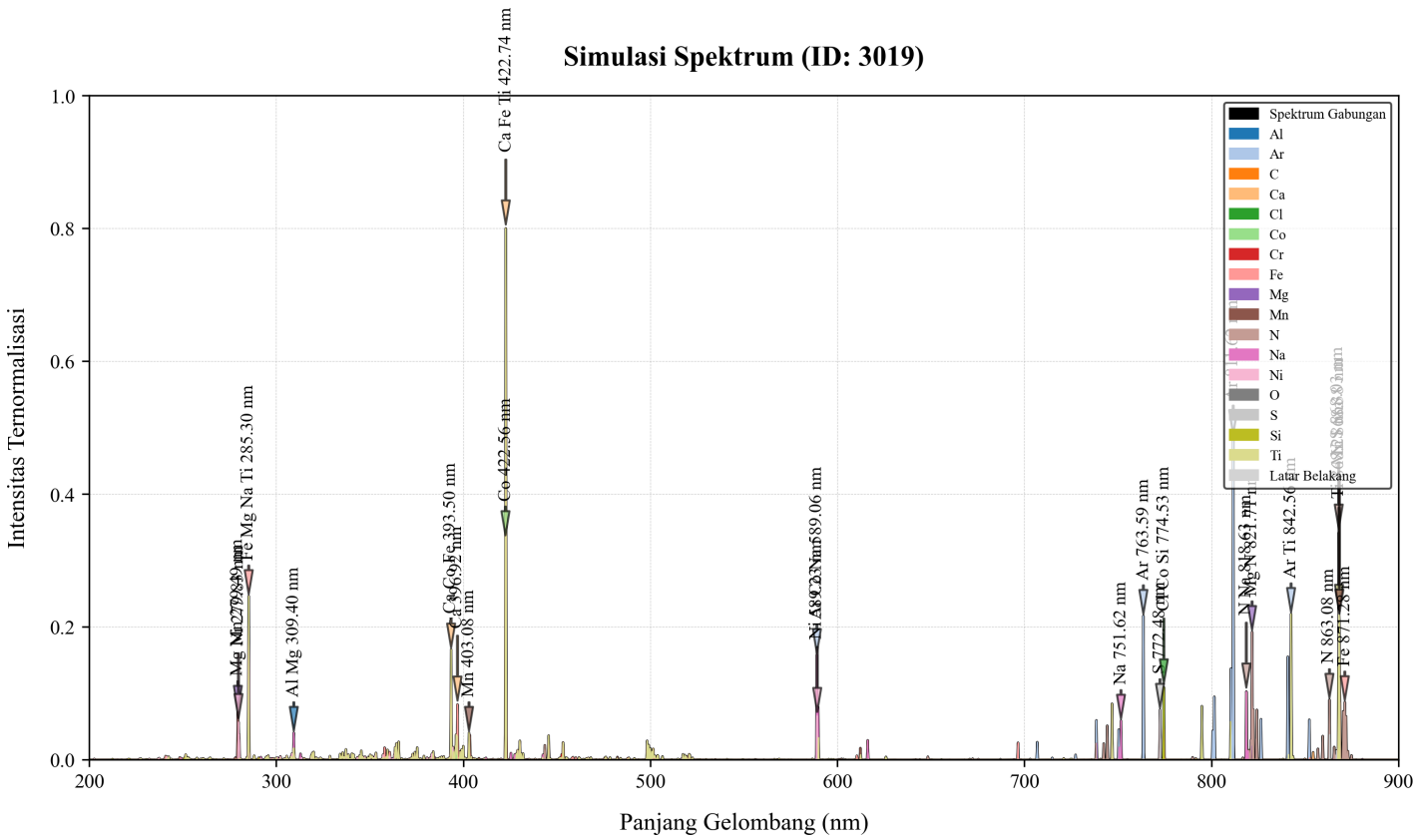

Figure 5: Spectral simulation plot for sample 3019 with multi-label encoding and detailed composition analysis.